Mohd Atif

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Summary

Data Scientist with 1 year of industry experience building production-grade ML pipelines, time-series forecasting models, and LLM-powered applications in Python, backed by hands-on backend engineering to ship them as scalable systems. Delivered a 20-35% improvement in forecast accuracy, 15% gross margin improvement, and 95% classification accuracy on a live deployed system. Skilled across the full data science lifecycle: data engineering, feature engineering, model development, validation, and deployment via FastAPI services and cloud infrastructure.

Professional Experience

Data Scientist Intern — *Arabania Indo-Arabic Restaurant*

June 2024 - May 2025, Lucknow

- Developed and deployed demand forecasting models using Python (statsmodels, scikit-learn) and time-series analysis, achieving 20% reduction in food waste and 12% procurement cost savings by accurately predicting daily inventory needs across 100K+ records.
- Designed and executed A/B pricing experiments end-to-end, applying hypothesis testing and statistical inference to identify a pricing strategy that improved gross margins by 15% within 2 months of deployment.
- Built supervised classification models for SKU performance prediction using feature engineering on behavioral and transactional data, identifying optimization opportunities that delivered a 12% operational improvement.
- Constructed automated Python and SQL analytics pipelines (CTEs, window functions) processing 100K+ records to extract seasonality and demand patterns, reducing manual reporting time by 40%.

Projects

European Power Market Forecasting System — GitHub

Python, FastAPI, XGBoost, ENTSO-E API, LLM Integration, Prompt Engineering, PnL Simulation

- Built an end-to-end electricity price forecasting pipeline on real ENTSO-E European market data, engineering 12+ features (lag-1/24/168h, rolling volatility, calendar effects) and training an XGBoost model achieving 20-35% improvement over baseline, outperforming 3 alternative models evaluated during development.
- Architected modular backend services covering data ingestion, validation, feature engineering, model inference, and trading signal generation; integrated an LLM via API with structured prompt engineering to auto-generate trader briefs, eliminating 3+ hours of manual report writing per week across 5 recurring report types.
- Designed an 8-point automated QA pipeline (schema validation, anomaly detection, correlation checks, freshness monitoring) reducing data quality incidents to zero, and implemented a backtesting framework across 12+ months of historical market data.

AI Customer Churn Decision Intelligence Platform — Live App

Python, FastAPI, REST APIs, Scikit-learn, LLM API Integration, Streamlit

- Trained and deployed a Logistic Regression classifier achieving 95% accuracy on behavioral features (usage, support tickets, tenure), serving real-time inference to 100+ customers via a production FastAPI backend with batch CSV processing.
- Designed and deployed a production-ready FastAPI backend exposing REST APIs for customer analysis, scenario simulation, and real-time prediction through an interactive Streamlit application, with a provider-agnostic LLM abstraction layer and explainability layer surfacing top 3 churn drivers per prediction.
- Designed a What-If simulation framework enabling stakeholders to evaluate intervention ROI independently; implemented environment-based configuration and cloud deployment on Render and Streamlit Cloud.

Real-Time Urban Analytics Dashboard — Live App

Python, REST APIs, Streamlit, PyDeck, ARIMA Forecasting, Anomaly Detection, Git

- Architected an automated multi-source data pipeline integrating 3 live REST APIs (traffic, weather, air quality) with schema validation, error handling, and 3-minute refresh caching, maintaining 99% data availability across continuous downstream modelling.
- Built statistical forecasting (ARIMA-style) and anomaly detection modules powering dashboards with real-time alerts and multi-city comparison across 5+ operational metrics.

Technical Skills

- **ML and Statistics:** XGBoost, scikit-learn, Logistic Regression, time-series forecasting, feature engineering, cross-validation, A/B testing, anomaly detection, backtesting, PnL simulation, statistical inference
- **Backend Engineering:** Python, FastAPI, REST APIs, SQL, database design, service integration, Streamlit
- **Cloud & DevOps:** Docker, AWS (EC2, Lambda), Kubernetes, CI/CD fundamentals, Render, Streamlit Cloud
- **LLMs and Agents:** Programmatic LLM API integration (OpenAI, Anthropic, Groq), LangChain, RAG pipelines, custom agent development, prompt engineering, agentic workflows, structured output validation
- **Programming and Data:** Python (Pandas, NumPy, statsmodels), SQL (CTEs, window functions, query optimisation), R, Git, automated data pipelines
- **Visualisation and Tools:** Power BI, Tableau, Matplotlib, Seaborn, Jupyter, data documentation

Certifications

- Google Data Analytics Professional Certificate (Foundations, Data Exploration, Data-Driven Decisions)

Education

Indian Institute of Technology, Madras — *2022-2025*

B.S. in Data Science and Applications